REP-027 · Balance redesign: fruit identity, strategy depth and pacing (proposal, no rule change)

Game	Fruit Market (rules version 2, `main` at 66bc299)
Report date	2026-10-27 09:04 (Asia/Kuala_Lumpur)
Type	Analysis, experiments and a patch plan. **Production rules unchanged.** All variants ran in a separate lab copy of the engine
Builds on	REP-025 (balance and pacing audit)
Proposed plan	PLAN-012 "Market rebalance: tighter town, fair Handcart, better apples" (awaiting approval)
Approval status	Awaiting approval (PLAN-012). This report approves nothing

How to read the numbers. Three kinds of statement appear below, and they are labelled:

- **[Code]**: a fact read from the source code.
- **[Sim]**: a measured result from computer players. The ± figures are 95% intervals. With 600 games per cell, a win rate is good to about ±4 points.
- **[Hyp]**: a hypothesis about how people will feel. Only the playtest (section 7) can confirm it.

Computer players are not people. Win rates show the shape of the game, not human results.

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1. Diagnosis: what the strategic game is today

The core loop (code). Each day you buy up to 4 fruits (5 with a Handcart), then set sealed prices per customer. The cheapest offer sells first, and freshness caps your price and breaks ties. The goal is the most cash at the end.

What decides games (sim, current rules, 3 players, 600 games per plan on held-out seeds):

1. **Two plans dominate.**
   - **Undercut**: always price $1 under the expected rival. It wins 43% (fair share is 33%).
   - **Salvage**: dump old stock cheaply and keep buying fresh. It wins 51%.
   - In 4 players it is worse: 45% and 54% (fair share 25%).
   - **Premium pricing** (hold out for top price) wins only 13% in 3 players and 6% in 4 players.
   - In short, cutting the price is almost always right. That is the opposite of an interesting sealed-bid game.
2. **The market is loose early, so price wars pay.**
   - Customer demand starts at 40% of what the shops can buy, while farmers bring 100%.
   - Shops are therefore fighting over too few customers with too much fruit.
   - 11% of 3-player sales and 16% of 4-player sales happen at $2 or less ("price-war share").

3. **The Handcart (and Big Basket) grows the market for everyone.** [Code] Demand and deliveries are sized from the sum of every shop's buy limit (`buyCapacity()` in `src/engine/game.ts`). So when one shop buys a Handcart, the town gets more customers and more fruit for all shops. That rewards whoever floods the market and blurs what the upgrade is for.
4. **Apples are a trap.**
   - Apples cost $4 fresh but are priced $1 below other fruit (`PRICE_SHIFT` in `data.ts`).
   - [Sim] Net profit per apple bought is $1.64, the lowest of the four fruits (mango is $2.58).
   - 25% of okay-age apples are still unsold at the end.
   - The "safe, long-keeping" identity does not pay, because almost nothing is held long enough for keeping to matter.
5. **Fruit is barely held, so decay rarely bites.**
   - [Sim] The average fruit sells 0.6 days after purchase, and 87–93% of fresh fruit sells the same or next day.
   - The only real "hold" decision is green mango (held 2.6 days, net $3.33 each) and green watermelon.
   - Spoilage is 1.2 fruits per shop per game in 3 players.
6. **Early leaders snowball.**
   - [Sim] The mid-game leader wins 57% in 3 players (fair is 33%).
   - Only 15% of 3-player games and 8% of 4-player games are won by whoever was last at mid-game.
   - 45% of games are blowouts.
7. **4 players is squeezed.** [Sim] Average final cash is $130 in 2 players, $80 in 3 and $56 in 4. Four-player shops barely grow, and price wars are most common there.

Summary. The tension the game wants is "freshness and timing against price". Today it collapses into "price lowest, buy fresh, repeat". Fruit identities exist on paper, but only mango's shows up in play.

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2. Fruit identity and decay

2a. Before and after (exact values)

Decay is unchanged in the recommendation. The reason is in 2c.

Fruit	Track, oldest → freshest (price drop $)	Cost: fresh / okay / not so / unripe (now → proposed)	Price shift on cards	Unripe nights	Intended identity
🫐 Blueberry	−4 · −2 · 0 (unchanged)	2/1/1/– (unchanged)	0	none	Cheap, fast, sell today
🍉 Watermelon	−3 · −1 · −1 · 0 · 0 (unchanged)	3/2/1/1 (unchanged)	0	1	Steady middle; small ripening bet
🍎 Apple	−2 · −1 · −1 · −1 · 0 · 0 · 0 (unchanged)	**4/3/1/1 → 3/2/1/1**	−1 (unchanged)	none	The keeper: safe margin any day, sells late

Fruit	Track, oldest → freshest (price drop $)	Cost: fresh / okay / not so / unripe (now → proposed)	Price shift on cards	Unripe nights	Intended identity
🥭 Mango	−3 · −1 · 0 · 0 (unchanged)	4/2/1/2 (unchanged)	+2	2	Premium; buy green and wait

Measured fruit economics (sim, 3 players, held-out seeds; "net" = sale price minus cost, per fruit bought):

Fruit	Net now	Net with package	Share of fruit bought, now → package	Note
Blueberry	$2.14	$3.43	26% → 24%	Still the quick-profit fruit
Watermelon	$1.90	$2.55	22% → 24%	Green watermelon nets $3.57
Apple	**$1.64 (last)**	**$2.69**	22% → 26%	Moves from trap to solid choice
Mango	$2.58	$4.26	30% → 25%	Green mango nets $4.07, held 2.9 days

All fruits gain under the package because the tighter market raises prices (section 3). What matters is the order: apple goes from clearly worst to level with watermelon.

2b. Specific investigations

Question	Test	Result	Verdict
Is apple too expensive?	Apple cost 3/2/1 (A1); apple price +$1 (A2); both (A3)	A1 and A3 both lift apple to mid-table. A3 adds cash and widens the Hard-vs-Normal gap slightly. A1 is the smaller change	**A1 recommended**
Should apples keep value even longer?	Apple drop −1 · −1 · 0…	No measurable change to any plan (all within ±3 points)	Rejected: no effect
Is blueberry too punishing?	Blueberry drop −3 · −1 · 0	No measurable change	Rejected: no effect
Is green mango too strong?	Green mango $2 → $3	Premium rises (27%) but Reserve becomes too strong (45%). The Hard-vs-Normal gap shrinks to 57% (3 players), so skill matters less	Optional experiment, not recommended now
Should unripe fruit stop arriving when it can't ripen before the end?	Unripe cutoff	No measurable change	Rejected
Does unsold fruit at the end need a salvage value?	$1 per fruit at the end; juice at the end	No measurable change	Rejected

2c. Why decay is not the lever

[Sim] Shops sell almost everything within a day, so changing decay curves moves nothing. Decay only matters if holding is a real choice. Holding becomes a real choice when the market is tight and

has gaps: some days your fruit is scarce and worth keeping, other days it is everywhere.

The package creates that through market size, not through faster rot. That honours "do not simply increase decay everywhere or force a spoilage quota". Spoilage actually falls (1.2 → 0.3 per shop) because shops buy more selectively.

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3. A deeper strategy layer

3a. The plans tested

Each plan is the Hard computer with one strategic lean layered on top (lab "styles").

Plan	Lean
Undercut	Always price $1 under the best rival guess
Salvage	Dump old stock cheaply, restock fresh
Premium	Keep prices near the top; accept unsold fruit
Reserve	Keep a cash cushion; buy only clear profits
Ripen	Prefer green mango and watermelon
Adaptive	Learn each rival's discount from past reveals
Turnover	Buy the cheapest stock and sell it fast
Event	Lean hard into event cards (festival, heatwave and so on)

3b. Matchup table: one plan against two (or three) standard Hard computers

Held-out seeds 100001–100600 (600 games per cell, about ±4 points). Fair share is 33% in 3 players and 25% in 4 players.

Plan	3p now	3p package	4p now	4p package
Undercut	**43**	25	**45**	21
Salvage	**51**	32	**54**	29
Premium	13	25	6	16
Reserve	30	36	–	–
Ripen	27	31	26	23
Adaptive (reads rivals)	36	34	–	–
Turnover	18	21	–	–
Event	32	30	–	–
Spread (best – worst)	**38 points**	**15 points**	**48 points**	**13 points**

Four different plans at one table (4 players, fair share 25%):

	Turnover	Reserve	Ripen	Event
Now	17	19	29	34
Package	16	33	32	20

Reading this. [Sim] Under the package no plan is a must-pick, and several different plans sit within a few points of fair. Turnover stays weakest. [Hyp] That is acceptable, because it is the "no thinking" plan. Reading rivals (Adaptive) stays strong, which keeps the sealed-price duel rewarding.

3c. Sealed-price analysis

- [Code] The cheapest offer sells first. Ties go to the fresher fruit, then Favoured Trade or Haggler, then buying order.
- [Sim] Price-war share (sales at $2 or less) falls from 11.4% to 7.0% in 3 players and from 16.3% to 7.6% in 4 players.
- [Sim] Sales within $1 of the customer's top price rise from 30% to 55%.
- **Meaning.** With fewer spare fruits around, the question changes from "how low must I go?" to "will they dare to go high, and is my fruit fresh enough to win a tie?". [Hyp] That is the more interesting guess.
- **Adaptive still wins 34%.** Reading rivals still beats autopilot.

3d. Handcart and Big Basket: the shared-market effect

- [Code] Today, any shop's Handcart or Big Basket raises everyone's demand and supply.
- **Proposed:** size the town from the base 4 fruits per shop, so upgrades help only their owner.
- [Sim] This one change does most of the work against undercutting (P1 test: Undercut 32 → 24, Reserve 32 → 39, comeback from last 23% → 28%).
- [Sim] Handcart is still bought in 99% of games, so it keeps its value.
- **Trade-off:** average cash falls about 12%. The demand start is raised (0.55 → 0.60) to give some back.

3e. Traits (supporting evidence; changes optional)

2,000 games per player count, current rules (development seeds) → package without the 0.60 demand start (held-out seeds).

Trait	3p now	3p package	4p now	4p package
Haggler	28.6	28.6	24.0	22.9
Big Basket	29.6	33.3	22.4	23.2
Bargain Hunter	38.4	36.8	29.9	29.5
Charmer	36.1	35.3	26.1	27.4
Local Favourite	30.0	31.5	18.4	20.5
Nest Egg	37.4	34.6	29.2	26.3
Spread	10 points	8 points	11.5 points	9 points

Optional trait tweak (Haggler wins ties before freshness; Local Favourite tip $3 → $4; Nest Egg $16 → $15):

- 3 players: spread 6 points (30.5–36.2).
- 4 players: 21.4–29.4.
- A small gain, kept as an optional experiment.
- Bargain Hunter needing a $3+ fruit and Big Basket also adding 3 shelf spaces: no measurable change.

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4. Pacing

4a. Standard game (10 days), held-out seeds

Measure	3p now	3p package	4p now	4p package
Average final cash	$80	$105	$56	$97
Day 1 income	$5.9	$8.8	$5.0	$9.2
Blowouts (winner far ahead)	45.5%	40.0%	42.5%	30.0%
Mid-game leader wins	56.8%	38.7%	47.8%	34.8%
Last at mid-game but wins	15.3%	28.8%	8.2%	18.3%
Lead changes per game	3.6	4.3	4.3	4.7
Close games	10.8%	15.2%	14.5%	15.8%

In 2 players, final cash goes from $130 to $146. A 4-player game is no longer the poor relation ($97 against $56); the 15% extra 4-player demand is what fixes this.

4b. Quick (8 days) and Long (14 days), 3 players

"Now" uses development seeds; "package" uses held-out seeds.

Measure	8d now	8d package	14d now	14d package
Undercut	45	29	40	18
Salvage	52	39	43	23
Premium	10.5	21	16.5	35
Reserve	–	34	–	41
4-plan table T/R/Ri/E	22/13/26/38	20/30/24/26	15/25/36/24	**10/45/35/10**
Comeback from last	19%	28%	17%	30%

- **Quick (8 days) becomes the most balanced length.**
- **Long (14 days) swings towards patient plans** (Reserve and Ripen); Turnover and Event fall to 10%.
- [Hyp] Patience winning a long game may feel right, but it is a watch item. Long-mode tuning is listed as an optional follow-up rather than mixed into this package.

4c. Provisional success criteria (computer-only, 3 and 4 players, held-out seeds)

#	Criterion	Target	Package result
S1	Against standard Hard computers, no single plan wins more than 1.25 × fair share (42% in 3p, 31% in 4p)	≤ 42% / ≤ 31%	36% / 29%: **PASS**
S2	At least 4 of 8 plans within ±7 points of fair (3p)	≥ 4	5: **PASS**
S3	Hard beats Normal head to head (skill still matters)	≥ 65%	66%: **PASS (narrow)**
S4	Comeback from last at mid-game	≥ 20% (3p), ≥ 15% (4p)	28.8% / 18.3%: **PASS**
S5	4-player cash at least 60% of 2-player cash	≥ 60%	66%: **PASS**
S6	Price-war share	≤ 10%	7.0% / 7.6%: **PASS**
S7	Long mode: no plan above 42% in the 4-plan table	≤ 42%	45% (Reserve): **FAIL**, watch item
S8	Human enjoyment: clearer choices, "I could have won" feeling	Playtest	**NOT RUN** (needs section 7)

5. Test method and evidence

- **Engine copy with switches.**
  - The lab is a copy of the rules engine with switches for each proposed change.
  - With every switch off it reproduces production exactly (checked on 450 games, identical results).
  - The production code was not touched.
- **Paired seeds and seat rotation.** Every variant replays the same seeds. Seats rotate each game, so every plan plays every seat equally often.
- **Many opponent types.**
  - The plans in 3a, plus Normal and Easy computers.
  - Crude single-rule bots, which all lose under both rule sets (0–3% wins), so they are not reported.
- **Sweeps.**
  - Eight runs (a baseline, a plan check and six sweeps): 813 test cells and 384,200 simulated games.
  - Round 1: demand × supply grid.
  - Rounds 2–3: market candidates, apple and mango costs, delivery forecast, end-of-game salvage.
  - Round 4: traits, Handcart basis, broke credit, earlier walk-ins, 4-player demand, Slow Season, unripe cutoff.
  - Round 5: decay variants, trait sample of 2,000 games, 8 and 14 days.
  - Round 6: final candidates on held-out seeds.
- **Validation.**

    - Seeds 1–600 were used to develop the package.
    - Seeds 100001–100600 (and 100001–102000 for traits) were kept back for the final check.
    - The current-rules numbers on held-out seeds match the development numbers within the error margin (Undercut 46 vs 43, Salvage 54 vs 51), so the results are not seed-specific.

- **Market League check** (300 games per stage, a computer standing in for you):

Stand-in skill	Village	Harbour	Old Town	Spice Lane	Grand Bazaar	Royal Market	Daily win
Normal, now	65	67	54	38	11	5	8
Normal, package	54	41	49	36	19	14	15
Hard, now	87	84	84	80	54	28	41
Hard, package	75	58	65	62	45	24	33

  The late-stage cliff softens, but early stages get harder because Easy and Normal computers do relatively better in a tight market. **Risk:** new players may find Harbour Steps harder. The league fix already suggested in REP-025 should follow this package, and the playtest checks it.

- **Rejected or no measurable effect:**
    - credit for broke shops;
    - earlier second walk-in;
    - Slow Season at −$1;
    - Bargain Hunter minimum cost;
    - Big Basket shelf bonus;
    - unripe cutoff;
    - end-of-game salvage or juice;
    - apple and blueberry decay changes.

- **Optional, not in the package:**
    - **Showing tomorrow's farmer delivery.** Undercut 28%, Turnover 14%, Hard-vs-Normal gap up to 72% in 2 players. It is an information feature with UI cost.
    - **Green mango at $3.**

- **Coverage gaps (NOT RUN):**
    - human play;
    - online rooms under new rules (needs verify:full after the patch);
    - Juice Press and Cool Room rework (Juice Press stays near-useless: 0.24 juiced per game);
    - favour-card balance;
    - every event card individually;
    - 2-player plan matchups (only mirror and skill tests);
    - Easy-level opponents in the plan tests.

- **Reproducibility.**
    - The lab, experiment lists and raw results are archived as `docs/workflow/reports/img/REP-027-lab.tar.gz`, with a README inside.

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6. Recommendation: one cohesive package (PLAN-012)

6a. Essential changes

For each change: what decision it improves, why it improves that decision, and what it could break.

#	Change	Decision improved	Why	What could break
E1	**Tighter town.** Demand starts at 60% of base capacity instead of 40% (growth unchanged at +45%). Deliveries are 80% + 20% growth instead of 100% + 50%	"How low do I price?" becomes "how high dare I go?"	Fewer spare fruits, so cutting is no longer automatic; premium and reserve become real plans	Higher cash totals make the achievement thresholds easier; new players see fewer bargains at the farmers market
E2	**Town size uses the base 4 fruits per shop.** Handcart and Big Basket no longer grow the market for everyone	"Should I buy a Handcart?" is about my own capacity, not flooding the town	Removes a hidden subsidy to market-flooding play; the main reason undercutting fell	Cash about 12% lower than E1 alone (offset by E1's 0.60 start); Handcart is slightly less exciting
E3	**4-player towns have 15% more demand**	Whether to fight or share in big games	4-player shops were starved ($56 average)	If over-tuned, 4 players becomes easier than 3 (now $97 against $105: fine)
E4	**Apple cost 4/3/1 → 3/2/1**	"Which fruit fills my last slot?"	Apple becomes a real safe choice instead of a trap ($1.64 → $2.69 net per fruit)	Apple is now equal to watermelon on cost; identity rests on keeping a week, which the text must say
E5	**RULES_VERSION 2 → 3** and compatibility work	–	Required by the rules-change policy	Old saves and old clients are cut off (by design, with clear messages)

6b. Optional experiments (not in PLAN-012)

1. Trait tweak (Haggler before freshness, Favourite $4, Nest Egg $15).
2. Green mango $3.
3. Show tomorrow's delivery.
4. Long-mode tuning (demand growth for 14 days).
5. Juice Press and Cool Room rework.
6. League stage rebalance (follows REP-025).

6c. Exact patch plan

File	Change
`src/engine/data.ts`	Add `export const MARKET = { demandStart: 0.6, demandGrowth: 0.45, supplyStart: 0.8, supplyGrowth: 0.2, fourPlayerDemand: 1.15 }`. Apple `cost: { fresh: 3, okay: 2, notso: 1, unripe: 1 }`. Apple `theme`: "Keeps a week and barely loses value. Never the best price, never a loss." Update the header comment (apple no longer "costs the most")
`src/engine/game.ts` (lines 179–209)	New `marketCapacity(s) = s.players.length * BASE_BUY` used by `demandScale` and `supplyScale` instead of `buyCapacity`. `demandScale`: `marketCapacity(s) * (MARKET.demandStart + MARKET.demandGrowth * progress) * (n === 4 ? MARKET.fourPlayerDemand : 1)`. `supplyScale`: `marketCapacity(s) * (MARKET.supplyStart + MARKET.supplyGrowth * progress) / FARMER_AVG_QTY`. Remove the unused `buyCapacity` if nothing else uses it. Fix the stale comment at line 179 ("3 fruit each a day")
`src/engine/ai.ts`	No code change. The AI reads `demandScale`, `FRUIT` costs and prices at run time (lab parity confirmed). Re-measure Easy, Normal and Hard gaps after the patch
`src/net/protocol.ts`	`RULES_VERSION = 3`. Old clients get the existing "please update" message from `server/room.ts` (line 203) and `server/meta.ts` (line 70)
`src/ui/store.ts`	No code change: line 66 already discards a game in progress with "from an older version of the rules". Release notes say so
`server/worker.ts` (lines 107 and 110)	Daily board key `daily:\${date}` → `daily:\${date}:r\${RULES_VERSION}`, so release-day scores under old and new rules never share a board. Replays already use server rules (`verifyDaily`)
`src/engine/daily.ts`	No change (same seeds; the outcomes simply follow the new rules)
`RULES.md`	Fruit table (apple $3/$2/$1, new character line). "The town grows": orders and deliveries are sized for 4 fruits per shop, whatever upgrades or traits you have; 4-player towns have 15% more customers. Handcart row: "Buy up to 5 fruits a day instead of 4 (the town doesn't grow with it)." Fix the existing wrong price-cap example (it says okay apples lose $2 and not-so lose $4; the real figures are $1 and $2). Balance notes refreshed
`src/ui/HowTo.tsx`, `src/ui/tutorial.ts`	No apple cost or demand numbers found; re-read after the patch. Fruit lines come from `FRUIT.theme`
`src/engine/data.ts` `UPGRADES.cart.text` / `TRAITS.basket.text`	Add "(the town doesn't grow with it)" to the Handcart; Big Basket unchanged
`src/ui/progress.ts`	Keep "Fruit Tycoon" ($200) and "Market Day" ($40/day). They become easier; earned badges stay. Owner choice listed in section 8
`tests/`	New tests: demand and supply are the same with or without a Handcart; demand rises 15% at 4 players; apple cost values. Update any snapshots
Balance records	Re-run `npm run balance` and the plan checks on the patched engine; attach to the PLAN-012 verification report

Release: this would ship as a minor version (1.2.0 or 1.3.0) with player notes, for example "Markets are livelier: customers come sooner, and big undercuts pay less. Apples are cheaper. Games in progress from the old version can't be resumed."

7. Human playtest plan

- **Who.** 4–6 people. Mix newcomers and people who have played 3+ games. At least one session with 4 players on phones (pass-and-play or online).
- **Format.**
  - Each tester plays current rules (live game) and the package (a preview copy built from a branch), in alternating order.
  - Each set is 2 games: one Standard and one Quick or Long.
  - Testers are not told which version is which.
- **Questions after each game** (1–5 scale, plus one sentence):
  1. Did you have more than one sensible way to play?
  2. Did prices feel like a guessing duel or a race to the bottom?
  3. Did each fruit feel different? Which one did you avoid, and why?
  4. Did you feel you could still win in the last third?
  5. Was the game the right length?
  6. Was anything confusing?
- **Watch for:**
  - the price testers set relative to the top price;
  - whether anyone buys a Handcart and says why;
  - whether apples get bought;
  - league stage 2 difficulty for newcomers.
- **Pass mark.**
  - The package scores at least as well on questions 1, 2 and 4 for most testers.
  - No tester finds it more confusing.
  - Any FAIL goes back to the lab.
- **No telemetry.** Notes are written by hand. No new data collection, as instructed.

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8. Decisions for the owner

1. **Approve PLAN-012** (E1–E5 as above), or ask for changes.
2. Achievement thresholds: keep $200 and $40 (recommended, simplest), or raise them in the same plan.
3. Run the human playtest before or after building the preview copy (recommended: build the package on a branch preview, then playtest, then publish).